

Network Security Incident Analysis Report

1. Executive Summary

Analysis of the provided network logs indicates a potential compromise involving multiple internal systems communicating with two external IP addresses identified in the logs as known malicious IPs: **186.20.20.27** and **176.30.30.27**.

The internal systems use addresses in the **101.123.171.x** range and repeatedly establish outbound connections to the two external destinations using several protocols and ports. The repeated communication across numerous internal hosts suggests that multiple systems may have been compromised or are otherwise participating in coordinated communication with external malicious infrastructure.

The available logs demonstrate **suspicious outbound communication**, but they do not by themselves prove that the external actor is moving laterally between internal systems. Additional authentication, firewall, DNS, endpoint, and internal network-flow logs would be needed to establish the initial compromise and determine whether lateral movement occurred.

2. Suspected Malicious Infrastructure

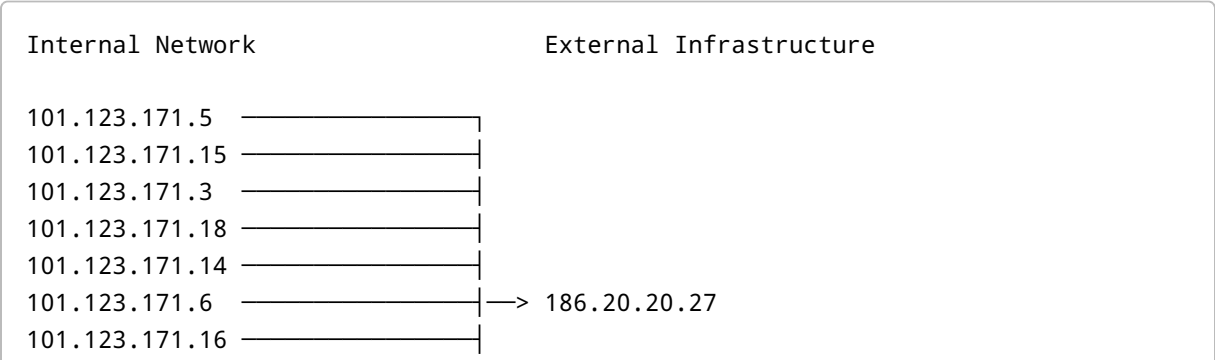
The two external IP addresses repeatedly contacted by internal systems are:

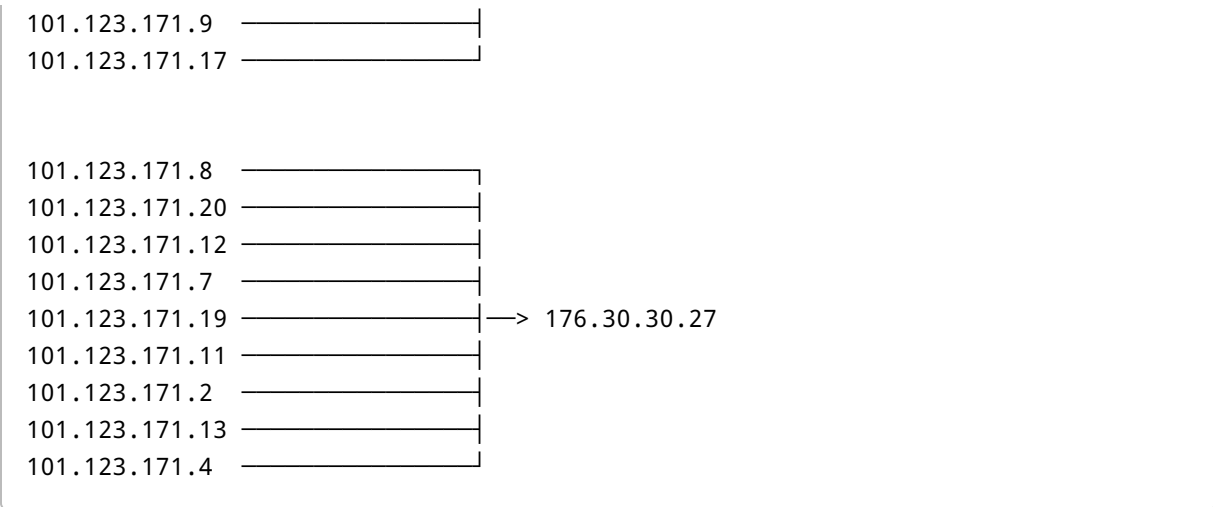
- **186.20.20.27**
- **176.30.30.27**

The log descriptions explicitly identify both destinations as **known malicious IP addresses**.

The internal `101.123.171.x` addresses should not automatically be considered the attacker. Instead, based on these logs, they appear to be **potentially compromised internal systems communicating with external infrastructure**.

Observed Communication Pattern





This pattern is consistent with multiple internal systems communicating with common external infrastructure.

3. Network Movement and Communication

The logs show communication traveling primarily in the following direction:

Internal systems → External malicious infrastructure

There is no direct evidence in the provided logs showing one internal system communicating with another internal system. Therefore, it would be inaccurate to state that the attacker is definitely performing lateral movement between the listed hosts.

Instead, the evidence suggests that multiple internal hosts may have been compromised and are independently communicating with the same external infrastructure.

This could potentially represent **command-and-control (C2) communication**, data transfer, or another form of malicious outbound communication.

4. Ports and Protocols Used

The suspected malicious infrastructure is being contacted through multiple network protocols and ports.

Protocol	Port	Potential Use	Observation
TCP	21	FTP	Repeated outbound connections
TCP	8080	Web/application traffic	Repeated abnormal/high-traffic connections

Protocol	Port	Potential Use	Observation
UDP	123	NTP	Unusual outbound communication
UDP	5000	Application-dependent	Repeated suspicious traffic
ICMP	N/A	Network communication/ diagnostics	Suspicious outbound ICMP

The use of multiple ports is notable because the traffic does not appear to be limited to a single application or communication channel.

TCP Port 21

TCP port **21** is traditionally associated with FTP. The logs describe high-volume outbound connections over this port.

Examples include:

- 101.123.171.5 → 186.20.20.27:21
- 101.123.171.7 → 176.30.30.27:21
- 101.123.171.16 → 186.20.20.27:21
- 101.123.171.4 → 176.30.30.27:21

TCP Port 8080

TCP **8080** is commonly used for web applications, proxies, and alternative HTTP services.

Examples include:

- 101.123.171.15 → 186.20.20.27:8080
- 101.123.171.12 → 176.30.30.27:8080
- 101.123.171.6 → 186.20.20.27:8080
- 101.123.171.13 → 176.30.30.27:8080

The logs specifically describe some of this traffic as having abnormal patterns or high traffic volumes.

UDP Port 123

UDP **123** is normally used by NTP for time synchronization.

Examples include:

- 101.123.171.3 → 186.20.20.27:123
- 101.123.171.19 → 176.30.30.27:123
- 101.123.171.9 → 186.20.20.27:123

While NTP itself is legitimate, communication with a destination identified as malicious makes these connections suspicious.

UDP Port 5000

UDP **5000** is not inherently malicious and can be used by various applications.

However, the repeated traffic to the known malicious IPs makes these connections suspicious.

Examples include:

- 101.123.171.8 → 176.30.30.27:5000
- 101.123.171.18 → 186.20.20.27:5000
- 101.123.171.11 → 176.30.30.27:5000
- 101.123.171.17 → 186.20.20.27:5000

ICMP

ICMP does not use TCP/UDP ports.

The logs show suspicious ICMP communication from:

- 101.123.171.20 → 176.30.30.27
- 101.123.171.14 → 186.20.20.27
- 101.123.171.2 → 176.30.30.27

The repeated ICMP communication is another indicator that the external infrastructure may be communicating with multiple compromised hosts.

5. Chronological Log Analysis

Time	Source	Destination	Protocol	Port	Assessment
14:30:00	101.123.171.5	186.20.20.27	TCP	21	High-volume outbound traffic
14:31:10	101.123.171.8	176.30.30.27	UDP	5000	Suspicious outbound traffic
15:45:22	101.123.171.15	186.20.20.27	TCP	8080	Abnormal traffic
16:20:45	101.123.171.20	176.30.30.27	ICMP	N/A	Suspicious outbound ICMP
16:35:00	101.123.171.3	186.20.20.27	UDP	123	Unusual outbound traffic
17:10:15	101.123.171.12	176.30.30.27	TCP	8080	High-volume outbound connection
17:45:30	101.123.171.18	186.20.20.27	UDP	5000	Suspicious outbound traffic
18:20:45	101.123.171.7	176.30.30.27	TCP	21	High-volume outbound connection

Time	Source	Destination	Protocol	Port	Assessment
18:35:00	101.123.171.14	186.20.20.27	ICMP	N/A	Suspicious ICMP
19:10:15	101.123.171.19	176.30.30.27	UDP	123	Unusual outbound traffic
19:45:30	101.123.171.6	186.20.20.27	TCP	8080	Abnormal traffic
20:20:45	101.123.171.11	176.30.30.27	UDP	5000	Suspicious outbound traffic
20:35:00	101.123.171.16	186.20.20.27	TCP	21	High-volume outbound traffic
21:10:15	101.123.171.2	176.30.30.27	ICMP	N/A	Suspicious ICMP
21:45:30	101.123.171.9	186.20.20.27	UDP	123	Unusual outbound traffic
22:20:45	101.123.171.13	176.30.30.27	TCP	8080	High-volume outbound traffic
22:35:00	101.123.171.17	186.20.20.27	UDP	5000	Suspicious outbound traffic
23:10:15	101.123.171.4	176.30.30.27	TCP	21	High-volume outbound traffic

6. Indicators of Compromise

External Indicators

186.20.20.27

- Repeated communication from multiple internal systems
- TCP/21
- TCP/8080
- UDP/123
- UDP/5000
- ICMP
- Identified in the logs as a known malicious IP

176.30.30.27

- Repeated communication from multiple internal systems
- TCP/21
- TCP/8080
- UDP/123
- UDP/5000
- ICMP
- Identified in the logs as a known malicious IP

Internal Indicators

Multiple hosts in the `101.123.171.x` range show suspicious outbound communication.

Affected addresses observed in the logs include:

101.123.171.2, .3, .4, .5, .6, .7, .8, .9, .11, .12, .13, .14, .15, .16, .17, .18, .19, and .20.

7. Assessment

The evidence indicates a **potential coordinated compromise involving multiple internal systems**.

The strongest indicator is the repeated communication between numerous internal hosts and the same two external IP addresses that the logs identify as malicious.

The traffic occurs across multiple protocols and ports, including TCP, UDP, and ICMP. The repeated connections over several hours suggest that this is unlikely to be a single isolated network event.

However, the provided logs do **not** establish:

1. Which internal system was initially compromised.
 2. How the initial compromise occurred.
 3. Whether the attacker actually moved laterally between internal systems.
 4. What data, if any, was accessed or exfiltrated.
 5. Whether the external IP addresses belong to the original attacker or are compromised third-party infrastructure.
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8. Recommended Investigation

The affected internal systems should be investigated for signs of compromise, beginning with the hosts showing the earliest suspicious communication.

Recommended evidence to collect includes:

- Firewall logs
- DNS query logs
- Authentication logs
- Windows Event Logs
- Linux authentication/system logs
- Endpoint detection and response (EDR) data
- Process and network connection information
- Internal network-flow records
- Proxy logs
- File creation/modification timestamps
- Persistence mechanisms such as scheduled tasks and services

The two external IP addresses should also be blocked or otherwise contained according to the organization's incident-response procedures while the investigation is conducted.

9. Conclusion

Based on the provided logs, **186.20.20.27 and 176.30.30.27 are the primary suspected malicious external IP addresses.**

The observed traffic follows the pattern:

Potentially compromised internal hosts → malicious external infrastructure

The external infrastructure is being contacted through **TCP ports 21 and 8080, UDP ports 123 and 5000, and ICMP.**

The large number of internal systems communicating with the same external destinations raises significant concern that multiple hosts may have been compromised. However, the available evidence is insufficient to conclusively demonstrate internal lateral movement. Further investigation is required to determine the initial point of compromise, the attacker's method of access, and whether sensitive information was accessed or exfiltrated.